

KIDNEY CROSS SECTION

This cutaway shows the kidney's main layers: the cortex and the medulla, which forms segments known as renal pyramids. The renal artery and vein circulate huge amounts of blood – about 1.2 litres per min ($2\frac{1}{2}$ pints/min) at rest, which is up to one-quarter of the heart's total output.

Renal cortex

Outer region of kidney; packed with microscopic structures called glomeruli, which make it look granular

Renal medulla

Region of capillaries around loops of tubules

Renal artery

Supplies blood to kidney; branches from the aorta

Renal vein

Removes cleaned blood, which then drains into inferior vena cava

Renal hilum

Junction where renal blood vessels and ureter pass into kidney

Renal column

Tissue between renal pyramids

Renal pyramid

Cone-shaped region of renal medulla

Major calyx

Several minor calyces merge to form a major calyx

Minor calyx

Cup-like cavity that receives urine from the renal papilla

Renal pelvis

Funnel-shaped tube into which the major calyces merge

Renal capsule

Thin covering of white fibrous tissue around the whole kidney

Renal papilla

Apex of the renal pyramid

Arcuate arteries and veins

Vessels forming arch-like links between the cortex and medulla

Interlobular arteries and veins

Branches of the renal artery and vein

Ureter

Tube for urine to pass to the bladder